

PRAKTIKUM SISTEM OPERASI

MODUL 10

SYSCALL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

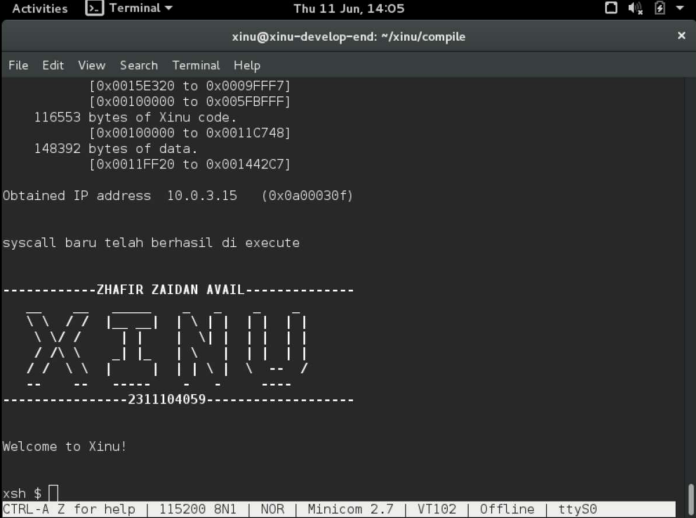
UNIVERSITAS TELKOM

2026

JURNAL/TP

1. Soal 1

[50 Poin] Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5!
(sertakan Screenshot kode dan hasil run)



```
xinu@xinu-develop-end: ~/xinu/compile
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
116553 bytes of Xinu code.
[0x00100000 to 0x0011C748]
148392 bytes of data.
[0x0011FF20 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

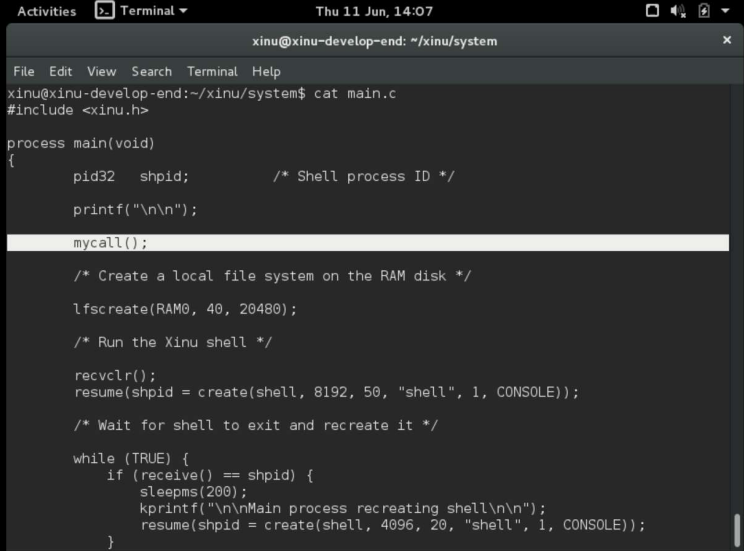
syscall baru telah berhasil di execute

-----ZHAFIR ZAIDAN AVAIL-----
XINU
-----2311104059-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Code main.c:



```
xinu@xinu-develop-end: ~/xinu/system
File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu/system$ cat main.c
#include <xinu.h>

process main(void)
{
    pid32  shpid;          /* Shell process ID */

    printf("\n\n");

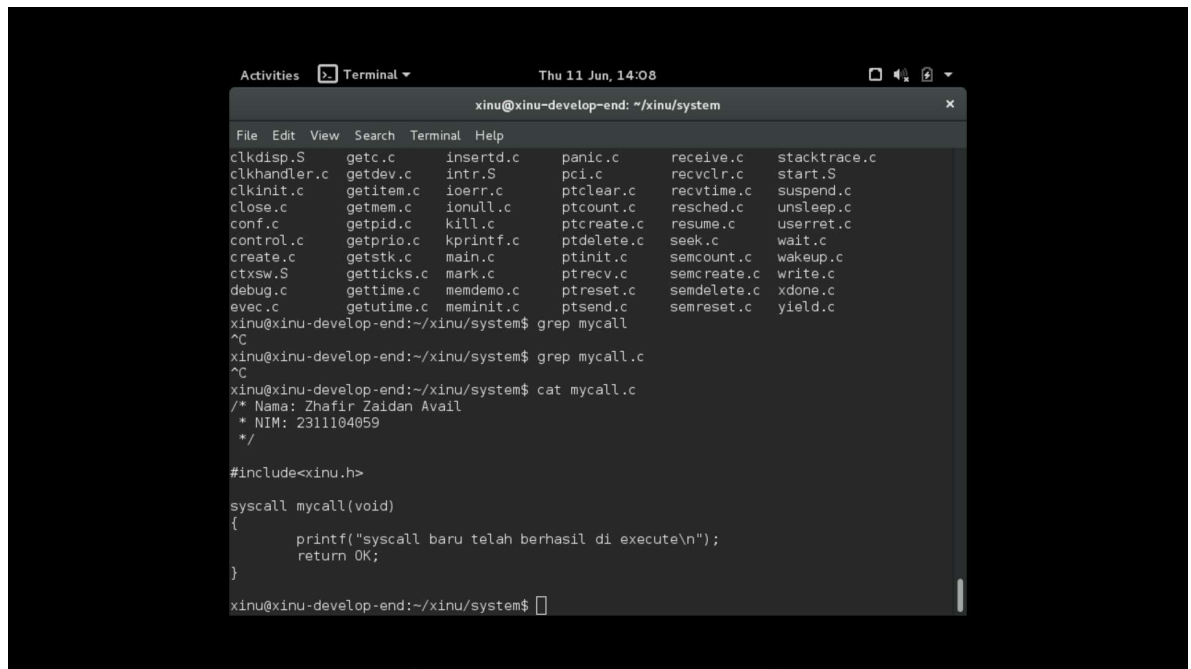
    mycall();

    /* Create a local file system on the RAM disk */
    lfscree(RAM0, 40, 20480);

    /* Run the Xinu shell */
    recvclr();
    resume(shpid = create(shell, 8192, 50, "shell", 1, CONSOLE));

    /* Wait for shell to exit and recreate it */
    while (TRUE) {
        if (receive() == shpid) {
            sleeps(200);
            kprintf("\n\nMain process recreating shell\n\n");
            resume(shpid = create(shell, 4096, 20, "shell", 1, CONSOLE));
        }
    }
}
```

Code mycall:



A terminal window titled "xinu@xinu-develop-end: ~/xinu/system" showing a directory listing of files in the xinu/system directory. The files are arranged in a grid. The user then runs the command `grep mycall`, which returns no results. Next, the user runs `cat mycall.c`, displaying the contents of the file. The file starts with a comment block containing the author's name and NIM, followed by an include statement for `xinu.h` and a function definition for `syscall mycall(void)` that prints a message and returns OK.

```
xinu@xinu-develop-end: ~/xinu/system
File Edit View Search Terminal Help
clkdisp.S      getc.c         insertd.c      panic.c        receive.c      stacktrace.c
clkhandler.c   getdev.c       intr.S         pci.c          recvclr.c     start.S
clkinit.c      getitem.c      ioerr.c        ptclean.c      recvtime.c    suspend.c
close.c        getmem.c       ionull.c       ptcoun.c       resched.c     unsleep.c
conf.c         getpid.c       kill.c         ptcreate.c     resume.c      userret.c
control.c      getprio.c      kprintf.c      ptdelete.c     seek.c        wait.c
create.c       getstk.c       main.c         ptinit.c       semcount.c    wakeup.c
ctxsw.S        getticks.c     mark.c         ptreceive.c    semcreate.c   write.c
debug.c        gettime.c      memdemo.c      ptreset.c      semdelete.c   xdone.c
evec.c         getutime.c     meminit.c      ptsend.c       semreset.c    yield.c

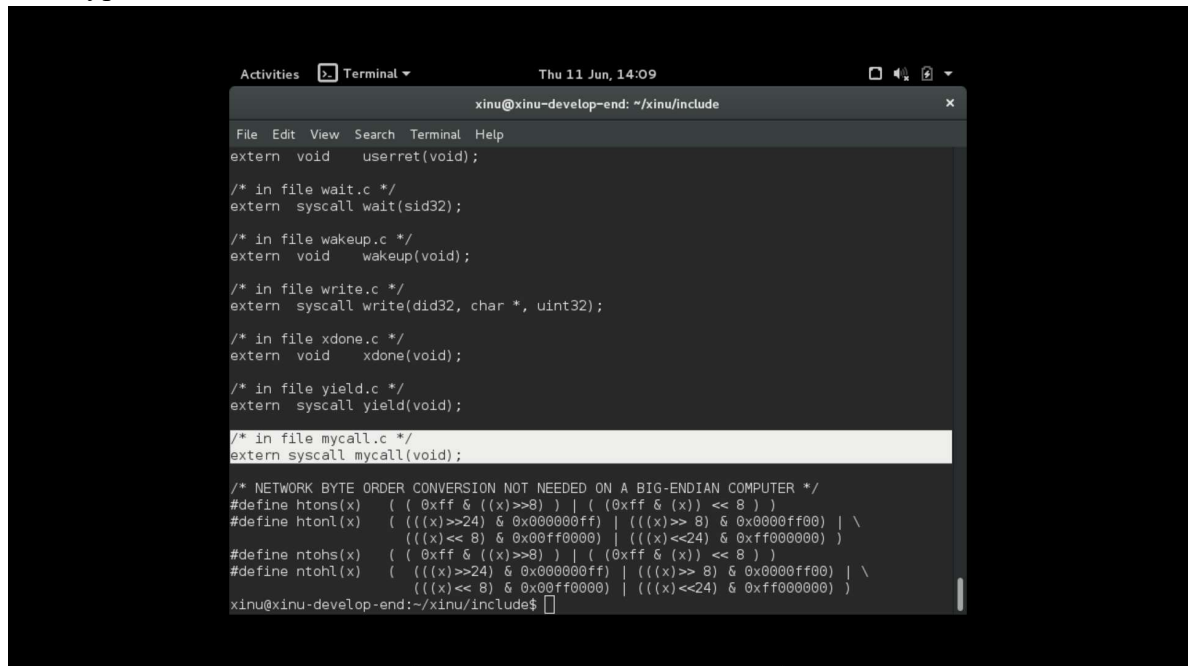
xinu@xinu-develop-end:~/xinu/system$ grep mycall
^C
xinu@xinu-develop-end:~/xinu/system$ grep mycall.c
^C
xinu@xinu-develop-end:~/xinu/system$ cat mycall.c
/* Nama: Zhaafir Zaidan Avail
 * NIM: 2311104059
 */

#include<xinu.h>

syscall mycall(void)
{
    printf("syscall baru telah berhasil di execute\n");
    return OK;
}

xinu@xinu-develop-end:~/xinu/system$
```

Prototypes.h:



A terminal window titled "xinu@xinu-develop-end: ~/xinu/include" showing the contents of a header file. The file contains several `extern` declarations for functions like `userret`, `wait`, `wakeup`, `write`, `xdone`, `yield`, and `mycall`. It also includes a section for network byte order conversion macros, with a comment indicating they are not needed on a big-endian computer.

```
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
extern void    userret(void);

/* in file wait.c */
extern syscall wait(sid32);

/* in file wakeup.c */
extern void    wakeup(void);

/* in file write.c */
extern syscall write(did32, char *, uint32);

/* in file xdone.c */
extern void    xdone(void);

/* in file yield.c */
extern syscall yield(void);

/* in file mycall.c */
extern syscall mycall(void);

/* NETWORK BYTE ORDER CONVERSION NOT NEEDED ON A BIG-ENDIAN COMPUTER */
#define htons(x) ( ((x)>>8) | ((x)<<8) )
#define htonl(x) ( (((x)>>24) & 0x000000ff) | (((x)>> 8) & 0x0000ff00) | \
  (((x)<< 8) & 0x00ff0000) | (((x)<<24) & 0xff000000) )
#define ntohs(x) ( ((x)>>8) | ((x)<<8) )
#define ntohl(x) ( (((x)>>24) & 0x000000ff) | (((x)>> 8) & 0x0000ff00) | \
  (((x)<< 8) & 0x00ff0000) | (((x)<<24) & 0xff000000) )

xinu@xinu-develop-end:~/xinu/include$
```

2. Soal 2

[25 Poin] Perbaiki syscall `chprio` (`xinu/system/chprio.c`) dengan memperhatikan validasi input

- Pastikan id adalah angka dari 0 – `NPROC` (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- `make clean`
- `make`

Code pada `chprio.c`:

```
xinu@xinu-develop-end: ~/xinu/system
File Edit View Search Terminal Help
)
pril6      newprio      /* New priority          */
)
{
    intmask mask;          /* Saved interrupt mask    */
    struct procent *prptr; /* Ptr to process's table entry */
    pril6      oldprio;     /* Priority to return       */

    mask = disable();
    if ((pid < 0) || (pid >= NPROC)) {
        restore(mask);
        return SYSERR;
    }

    if (newprio <= 0) {
        restore(mask);
        return SYSERR;
    }

    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    restore(mask);
    return 0xffff & oldprio;
}
```

Hasil makefile:

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh udpecho.o ../shell/xsh udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh udpserver.o ../shell/xsh udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh uptime.o ../shell/xsh uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.
xinu@xinu-develop-end:~/xinu/compile$
```

3. Soal 3

Lakukan hal-hal berikut ini

- Edit xsh_uptime.c
Tambahkan kode berikut

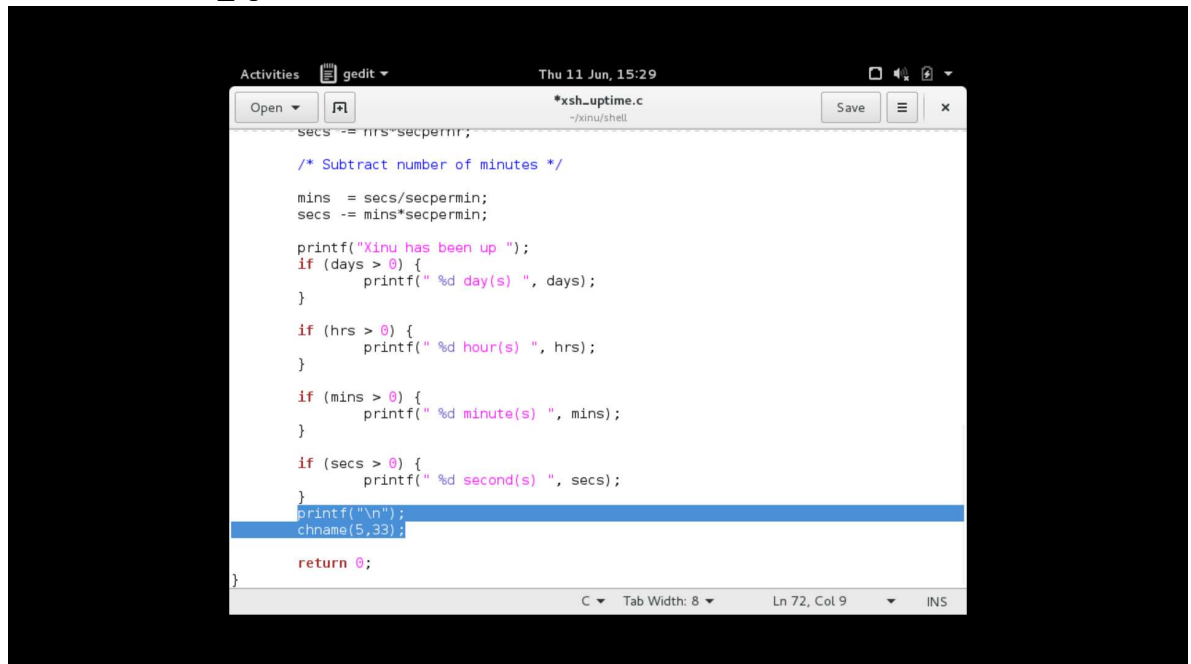
```
/* Check for valid number of arguments */  
  
if (nargs > 1) {  
    fprintf(stderr, "%s: too many arguments\n", args[0]);  
    fprintf(stderr, "Try '%s --help' for more information\n",  
            args[0]);  
    return 1;  
}  
  
// tambahan code  
chprio(5,33) // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5
```

- Compile source code tersebut dengan perintah
 - make clean
 - make
- Jalankan perintah ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5
- Jalankan uptime
 - xsh \$ uptime
 - Perhatikan hasil perintah tersebut
- Jalankan ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

[25 Poin] Testing chprio syscall yang telah diubah

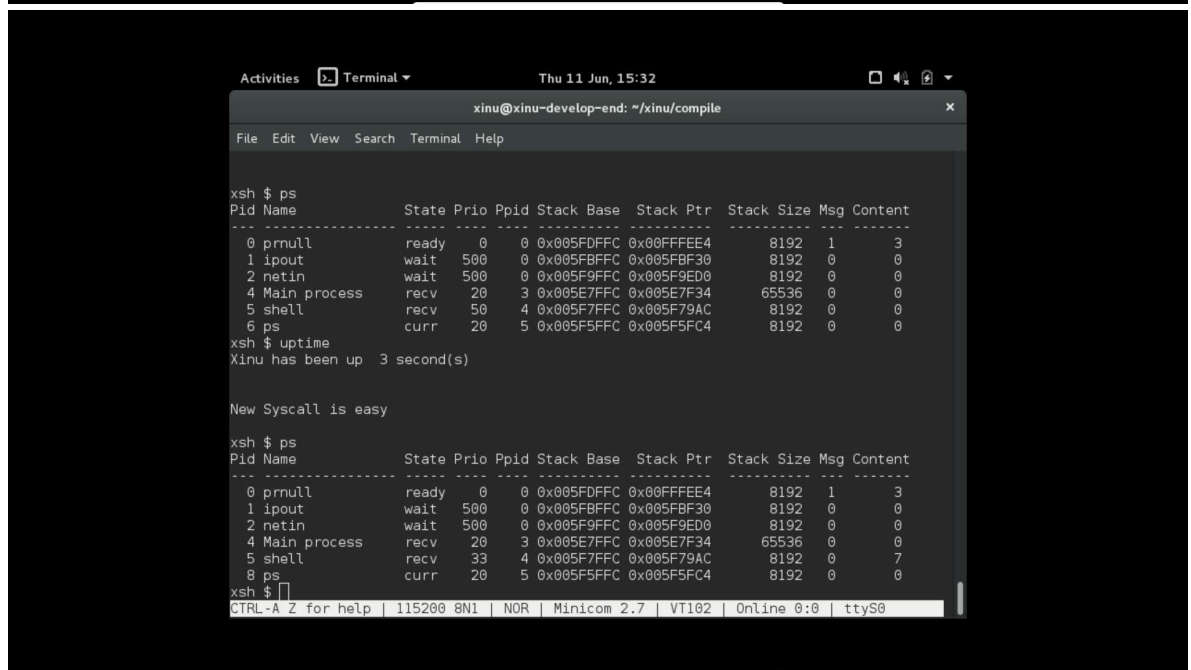
- Testing prioritas tidak boleh < 0 : Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c
- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”
- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)

tambahan di xhs_uptime.c:



The screenshot shows a gedit editor window titled "xhs_uptime.c" with the following C code:

```
secs -= hrs*secperrn;  
  
/* Subtract number of minutes */  
  
mins = secs/secperrn;  
secs -= mins*secperrn;  
  
printf("Xinu has been up ");  
if (days > 0) {  
    printf(" %d day(s) ", days);  
}  
  
if (hrs > 0) {  
    printf(" %d hour(s) ", hrs);  
}  
  
if (mins > 0) {  
    printf(" %d minute(s) ", mins);  
}  
  
if (secs > 0) {  
    printf(" %d second(s) ", secs);  
}  
printf("\n");  
chname(5,33);  
  
return 0;  
}
```



The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile". The user runs the command `ps`, which displays a table of processes. Then, the user runs `uptime`, which outputs "Xinu has been up 3 second(s)".

Output of `ps`:

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size	Msg	Content
0	prnull	ready	0	0	0x005F0FFC	0x00FFFFE4	8192	1	3
1	ipout	wait	500	0	0x005FBFFC	0x005FBF30	8192	0	0
2	netin	wait	500	0	0x005F9FFC	0x005F9ED0	8192	0	0
4	Main process	recv	20	3	0x005E7FFC	0x005E7F34	65536	0	0
5	shell	recv	50	4	0x005F7FFC	0x005F79AC	8192	0	0
6	ps	curr	20	5	0x005F5FFC	0x005F5FC4	8192	0	0

Output of `uptime`:

```
Xinu has been up 3 second(s)
```